COURSE SYLLABUS

AFS401: Agentforce Service Agent

Course Overview: This technical boot camp is designed for Salesforce Architects, Developers, and Technical Consultants. You will learn to design, build, test, and deploy enterprise-grade autonomous and hybrid AI agents using the **Salesforce Agentforce** platform and the **Agent Script** domain-specific language (DSL).

Throughout the course, you will build two production-ready agents for a multi-sided delivery platform ("Pronto"): a customer-facing Service Agent and a Merchant Support Specialist. The curriculum moves from visual canvas authoring to programmatic DSL editing, multi-modal grounding in Prompt Builder, Retrieval-Augmented Generation (RAG), and Omni-Channel deployments.

Curriculum Matrix

Lesson	Core Milestone	Key Capabilities Showcased
L1: Builder & Script	Deploy primary Service Agent	Agentforce Builder, Bi-directional sync, standard topic routers, multilingual support (`language:` block), trace logging.
L2: Actions & Guards	Deploy Merchant Agent via code	Agent Script (.agent DSL), compiler diagnostics, Apex/Flow actions, `available when` conditionals, explicit user confirmation.
L3: Prompt Builder	Automate reviews & outreach	Field Generation templates, Flex templates, Record & Related-List Grounding, Trust Layer masking, temperature tuning.
L4: Data 360 & RAG	Ingest Pronto Policy Pack	Agentforce Data Library, hybrid search (Vector + BM25), custom retrievers, metadata pre-filtering, `knowledge:` citations.
L5: Hybrid Reasoning	Implement OTP Identity Checks	Deterministic gates, mutable/linked variables, `before_reasoning`, finite-state flow transitions, step counters.
L6: Channel Deploy	Web & Omni-Channel rollout	Messaging for In-App/Web (MIAW), Omni-Flow routing, contextual human escalation, conversation transcript passing.
L7: Observability	Automated Regression Testing	Agentforce Testing Center, synthetic test generation, LLM-as-a-judge, Quality Score optimization, Intent classification drift.
Wrap: Demo Day	Production Showcase	End-to-end multi-agent evaluation, architectural defense, and security policy adherence validation.

Lesson 1: Agentforce Builder & Agent Script Fundamentals

Lesson 1 immerses you in the next-generation authoring environment of Salesforce Agentforce. You will build the primary customer-facing service agent, learning the dual-layer architecture of visual low-code canvas modeling and the high-code Agent Script DSL.

Concepts & Technical Topics Learned

- **Agentforce Builder Architecture:** Understand the 3 authoring modalities: Conversational Agent Assistant, Visual Canvas View, and Script View. Discover bi-directional synchronization and how canvas blocks compile directly to YAML-based Agent Script. Master the difference between *Simulate Mode* (dry-run reasoning) and *Live Test Mode* (executing real actions against sandbox records).
- **Agent Script DSL Essentials:** Learn the script file structure (`.agent` metadata): `system:`, `config:`, `variables:`, and `subagents:`. Master syntax, indentation rules, prompt assembly pipelines, and resource referencing (`@actions`).
- **Subagent Routing Architecture:** Explore the Atlas Reasoning Engine's hierarchical dispatch pattern, intent classification mechanics, and standard system subagents for intercepting off-topic or toxic utterances.
- **Localization & Observability:** Implement native localization using the `language:` block. Activate conversation recording, capture telemetry, and inspect the Agent Execution Trace to view raw LLM input/output payloads.

Lesson 2: Subagent Design & Tool Guardrails

You will build the Merchant Support Agent entirely from code using Agent Script. You will expand capabilities by wiring Invocable Apex and Flows, configuring critical enterprise safety controls that prevent unauthorized tool execution.

Concepts & Technical Topics Learned

- **Pro-Code Agent Authoring:** Develop agents via local IDEs utilizing Agent Script schema definitions. Utilize the diagnostic compiler to interpret linter warnings and unreachable paths.
- **Agent Action Architecture:** Wire standard Invocable Apex classes, Autolaunched Flows, and external APIs. Parameterize dynamic conversation context variables to action input/output primitives (`@inputs`, `@outputs`).
- **Deterministic Safety Controls:** Write `available when` Boolean guard expressions to ensure tools are only presented to the LLM when specific state conditions are met. Enforce `require_user_confirmation` (Human-in-the-Loop) before destructive or financial transactions. Establish semantic boundaries to eliminate tool-selection hallucinations.

Lesson 3: Prompt Builder & Context Grounding

Focus on engineering structured prompt templates in Prompt Builder. You will master grounding techniques—feeding real-time CRM context into foundation models—and evaluate how parameters like temperature and model selection impact deterministic business outputs.

Concepts & Technical Topics Learned

- **Prompt Template Typology:** Deploy Field Generation Templates for automated summarization directly on Lightning record pages, and Flex Templates for dynamic, multi-purpose tool calling inside Agentforce subagents.
- **Model Configuration & Trust Layer:** Calibrate hyperparameters (Temperature and Top-p). Leverage the Einstein Trust Layer for dynamic data masking (PII/PCI), toxicity scoring, zero-data retention, and full audit trailing.

Grounding Techniques Architecture

Grounding Mechanism	Execution Layer	Best Use Case
Record Merge Fields	Declarative	Direct field values on the active record (e.g., Status, Name).
Related-List Grounding	Declarative	Child object records (e.g., Recent Orders, Case History).
Flow Merge Fields	Low-Code Engine	Conditional queries, cross-object traversal, and logic calculations.
Apex Grounding	Pro-Code	External REST API data, complex aggregation, or large-scale data manipulation via `PromptGroundingProvider`.

> **Model Selection Gateway:** Depending on the complexity of your grounding framework and prompt instructions, you will test swapping models (e.g., OpenAI GPT-4o vs. Anthropic Claude) via the Trust Layer gateway to observe differing adherence to JSON output requirements.

Lesson 4: Unstructured Data & RAG (Data 360)

Connect Agentforce to unstructured knowledge bases using Data Cloud. You will ingest policy manuals, configure vector embeddings, build hybrid search indexes, and anchor agent responses directly to verified enterprise documentation to prevent hallucinations.

Concepts & Technical Topics Learned

- **Agentforce Data Library:** Ingest unstructured content (PDFs, HTML, Articles) into Data Spaces. Evaluate document chunking strategies, token overlaps, and Vector database embedding generation.
- **Hybrid Search Indexing:** Combine Dense Vector Search (semantic similarity) with Sparse Lexical Search (BM25 keyword matching for exact SKUs or policy numbers) using Reciprocal Rank Fusion.
- **Custom Retrievers & Grounding:** Configure custom retrievers with metadata pre-filtering (e.g., restricting searches to `Audience == 'External'`). Implement the `knowledge:` block in Agent Script to automate inline citation linking and evaluate Groundedness metrics.

Lesson 5: Hybrid Reasoning & Deterministic Gates

Pure LLM reasoning is inherently non-deterministic. Enterprise agents require strict adherence to regulatory rules and fraud controls. Learn to combine generative reasoning with finite-state machine (FSM) determinism using Agent Script control gates.

Concepts & Technical Topics Learned

- **Variable State Management:** Manage Mutable Variables (writable, session-scoped memory flags) and Linked Variables (read-only context bound to the runtime session).
- **Advanced Guardrails:** Use `filter_from_agent: true` to hide sensitive CRM outputs (e.g., OTP answers, fraud scores) from the LLM's context window while keeping them accessible to the deterministic code pipeline.
- **Execution Lifecycle Hooks:** Utilize the `before_reasoning` block for pre-flight security checks, and the `after_reasoning` block for deterministic post-processing. Force deterministic finite-state transitions to bypass LLM judgment when violation thresholds (like repeated failed logins) are breached.

The Hybrid Reasoning Execution Lifecycle:

1. `start_agent` : Intent classification to correct subagent.
2. `before_reasoning` : Pre-LLM deterministic actions (Fetch data, check security flags).
3. **Prompt Assembly**: Resolve variables, evaluate logic, inject RAG context.
4. **LLM Reasoning**: Select tools filtered by guard clauses.
5. `after_reasoning` : Post-LLM verification, counter incrementation, or forced escalation transitions.

Lesson 6: Omni-Channel Messaging Integration

Agents must live where customers are. Deploy your Agent across enterprise customer channels using Messaging for In-App and Web (MIAW), configure Omni-Channel routing, and manage bidirectional handoffs between AI and human agents.

Concepts & Technical Topics Learned

- **Omni-Channel Service Infrastructure:** Configure Service Channels, Presence Configurations, and Capacity Models. Establish fallback queues to ensure routing fail-safes when agents encounter system exceptions.
- **Messaging for In-App and Web (MIAW):** Deploy Embedded Service snippets, customize connection messaging, and map hidden pre-chat fields (browser locale, active session ID) into the agent's linked variables.
- **Omni-Flow Routing & Escalation:** Build routing flows that evaluate incoming messaging sessions before invoking the agent. Trigger `@utils.transfer_to_omnichannel` for seamless human handoffs.
- **Context Preservation:** Pass the full LLM conversation transcript, summary, and contextual intent to the Service Cloud Console so human agents do not ask repetitive questions.

> **Dynamic Language Preservation:** During the human handoff process, the system natively preserves the detected conversation language, ensuring the Omni-Channel routing engine targets an appropriately skilled human agent based on locale.

Lesson 7: Observability & The Testing Center

Deploying an agent is only the first step. Learn how to manage the AI lifecycle by running synthetic regression test suites at scale, analyzing conversation telemetry in Observe & Optimize, and systematically eliminating failure modes.

Concepts & Technical Topics Learned

- **Agentforce Testing Center Architecture:** Understand why traditional unit testing fails with conversational AI. Generate statistical test suites via CSV batch uploads, historical transcript imports, and AI-generated adversarial cases.
- **LLM-as-a-Judge Evaluation:** Automate scoring across standardized dimensions: Instruction Adherence, Topic Classification Accuracy, Action Selection Precision, and Groundedness.
- **Observe & Optimize Telemetry:** Track deflection rates and average handling time. Visualize semantic clustering of queries on the *Sessions & Intents Dashboard* to identify intent drift. Deep-dive into Quality Score diagnostics to locate prompt ambiguities.
- **Lifecycle Improvement:** Perform root-cause analysis of agent failures, distinguishing between instruction deficits, tool parameter mismatches, and RAG retrieval misses.

Course Wrap & Capstone: Agentforce Demo Day

The course concludes with a live presentation of your production agents. You will defend your design decisions, demonstrating that your agents are capable, robust, and aligned with enterprise security policies.

Deliverables & Evaluation Criteria

- **Functional Demonstration:** Live execution of multi-turn customer and merchant scenarios across web messaging channels.
- **Architectural Defense:** Explaining Agent Script structure, subagent boundaries, hybrid reasoning gates, and grounding choices.
- **Guardrail Proof:** Demonstrating that the agent successfully resists prompt injection and respects transaction safety constraints.
- **Testing Evidence:** Presenting the Agentforce Testing Center report showcasing test coverage and high quality scores.

Technical Concept Master Index

For quick reference, the following core Agentforce technologies and design patterns are integrated throughout this syllabus:

- **Authoring & Code:** Agentforce Builder, Canvas View, Script View, Agent Script DSL (`.agent`), Assistant conversational scaffolding, Agentforce DX.
- **Agent Architecture:** Atlas Reasoning Engine, `start_agent` (Router), Subagents (Topics), Mutable & Linked Variables, Session Memory.
- **Deterministic Logic:** Hybrid Reasoning, `before_reasoning` , `after_reasoning` , `available when` , `require_user_confirmation` , `filter_from_agent` , Finite-State Machine gates.
- **Grounding & Prompts:** Prompt Builder, Field Generation Templates, Flex Templates, Record Merge Fields, Related-List Grounding, Template-Triggered Prompt Flows, Apex Grounding (`PromptGroundingProvider`), Einstein Trust Layer.
- **RAG & Unstructured Data:** Data Cloud / Data 360, Agentforce Data Library, Chunking Strategies, Vector Embeddings, Hybrid Search Indexing (Dense Vector + BM25 Lexical), Custom Retrievers, Metadata Pre-filters, `knowledge:` block, Citation Engine.
- **Channels & Engagement:** Messaging for In-App and Web (MIAW), Enhanced Chat v2, Omni-Channel Routing, Omni-Flow, Fallback Queues, Contextual Escalation.
- **Evaluation & Operations:** Agentforce Testing Center, AI-Generated Test Suites, LLM-as-a-Judge, Observe & Optimize, Sessions & Intents Clustering, Quality Score, Session Tracing.